$$
\begin{bmatrix}
1-B_{x,\mu'} & 0 & -B_{x,\alpha'} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-B_{z,\mu'} & 0 & 1-B_{z,\alpha'} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & -\iota_{xy} & -\iota_{xz} & 0 & 0 & 0 & 0 & 0 \\
-B_{m,\mu'} & 0 & -B_{m,\alpha'} & -\iota_{yx} & 1 & -\iota_{yz} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\iota_{zx} & -\iota_{zy} & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\begin{Bmatrix}
\Delta\mu' \\ \Delta\beta' \\ \Delta\alpha' \\ \Delta\bar{p}' \\ \Delta\bar{q}' \\ \Delta\bar{r}' \\ \Delta\varsigma_x' \\ \Delta\varsigma_y' \\ \Delta\varsigma_z' \\ \Delta\phi' \\ \Delta\theta' \\ \Delta\psi'
\end{Bmatrix}
=
\begin{bmatrix}
0 & D_{x,\delta_e} & 0 \\
D_{y,\delta_a} & 0 & D_{y,\delta_r} \\
0 & D_{z,\delta_e} & 0 \\
D_{\ell,\delta_a} & 0 & D_{\ell,\delta_r} \\
0 & D_{m,\delta_e} & 0 \\
D_{n,\delta_a} & 0 & D_{n,\delta_r} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0 \\
0 & 0 & 0
\end{bmatrix}
\begin{Bmatrix}
\Delta\delta_a \\ \Delta\delta_e \\ \Delta\delta_r
\end{Bmatrix}
$$

$$
+
\begin{bmatrix}
A_{x,\mu} & A_g S_{\phi_o} C_{\theta_o} & A_{x,\alpha} - A_g T_{\phi_o} S_{\phi_o} C_{\theta_o} & 0 & A_{x,\tilde{q}} & 0 & 0\,0\,0 & 0 & -A_g C_{\theta_o} & 0 \\
-A_g S_{\phi_o} C_{\theta_o} & A_{y,\beta} & -A_g T_{\phi_o} S_{\theta_o} & A_{y,\bar{p}} & 0 & A_{y,\bar{r}}-1 & 0\,0\,0 & A_g C_{\phi_o} C_{\theta_o} & -A_g S_{\phi_o} S_{\theta_o} & 0 \\
A_{z,\mu}+A_g T_{\phi_o} S_{\phi_o} C_{\theta_o} & A_g T_{\phi_o} S_{\theta_o} & A_{z,\alpha} & 0 & A_{z,\tilde{q}}+1 & 0 & 0\,0\,0 & -A_g S_{\phi_o} C_{\theta_o} & -A_g C_{\phi_o} S_{\theta_o} & 0 \\
0 & A_{\ell,\beta} & A_{\ell,\alpha} & A_{\ell,\bar{p}}+\eta_{xx} & -\eta_{xy} & A_{\ell,\bar{r}}+\eta_{xz} & 0\,0\,0 & 0 & 0 & 0 \\
A_{m,\mu} & A_{m,\beta} & A_{m,\alpha} & \eta_{yx} & A_{m,\tilde{q}}+\eta_{yy} & -\eta_{yz} & 0\,0\,0 & 0 & 0 & 0 \\
0 & A_{n,\beta} & A_{n,\alpha} & A_{n,\bar{p}}-\eta_{zx} & \eta_{zy} & A_{n,\bar{r}}+\eta_{zz} & 0\,0\,0 & 0 & 0 & 0 \\
C_{\theta_o} & S_{\phi_o} S_{\theta_o} & C_{\phi_o} S_{\theta_o} & 0 & 0 & 0 & 0\,0\,0 & 0 & -S_{\theta_o} & 0 \\
0 & C_{\phi_o} & -S_{\phi_o} & 0 & 0 & 0 & 0\,0\,0 & 0 & 0 & C_{\theta_o} \\
-S_{\theta_o} & S_{\phi_o} C_{\theta_o} & C_{\phi_o} C_{\theta_o} & 0 & 0 & 0 & 0\,0\,0 & 0 & -C_{\theta_o} & 0 \\
0 & 0 & 0 & 1 & S_{\phi_o} T_{\theta_o} & C_{\phi_o} T_{\theta_o} & 0\,0\,0 & 0 & A_g T_{\phi_o}/C_{\theta_o} & 0 \\
0 & 0 & 0 & 0 & C_{\phi_o} & -S_{\phi_o} & 0\,0\,0 & -A_g T_{\phi_o} C_{\theta_o} & 0 & 0 \\
0 & 0 & 0 & 0 & S_{\phi_o}/C_{\theta_o} & C_{\phi_o}/C_{\theta_o} & 0\,0\,0 & 0 & A_g T_{\phi_o} T_{\theta_o} & 0
\end{bmatrix}
\begin{Bmatrix}
\Delta\mu \\ \Delta\beta \\ \Delta\alpha \\ \Delta\bar{p} \\ \Delta\bar{q} \\ \Delta\bar{r} \\ \Delta\varsigma_x \\ \Delta\varsigma_y \\ \Delta\varsigma_z \\ \Delta\phi \\ \Delta\theta \\ \Delta\psi
\end{Bmatrix}
\tag{9.8.20}
$$